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Key Points:
- A quantitative diatom-based ice-cover model has been created
- Inferred past ice-cover trends are similar to the regional climate history
- Diatoms can provide a valuable tool to reconstruct past ice-cover regimes

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Reconstructing lake ice cover in subarctic lakes using a diatom-based inference model

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Abstract A new quantitative diatom-based lake ice cover inference model was developed to reconstruct past ice cover histories and applied to four subarctic lakes. The used ice cover model is based on a calculated melting degree day value of +130 and a freezing degree day value of −30 for each lake. The reconstructed Holocene ice cover duration histories show similar trends to the independently reconstructed regional air temperature history. The ice cover duration was around 7 days shorter than the average ice cover duration during the warmer early Holocene (approximately 10 to 6.5 calibrated kyr B.P.) and around 3–5 days longer during the cool Little Ice Age (approximately 500 to 100 calibrated yr B.P.). Although the recent climate warming is represented by only 2–3 samples in the sediment series, these show a rising trend in the prolonged ice-free periods of up to 2 days. Diatom-based ice cover inference models can provide a powerful tool to reconstruct past ice cover histories in remote and sensitive areas where no measured data are available.

1. Introduction

Ice-covered freshwater bodies of various sizes represent a considerable fraction of the high-latitude sub-Arctic and tundra landscape. Projected climate change is expected to strongly alter freshwater ice regimes in cold regions [*Prowse et al.,* 2011c]. Climate influences the timing of lake ice melt and freeze onset, ice duration, and lake thermal dynamics that have feedback to the climate system initiating further changes in heat fluxes and radiation environment [e.g., *Rouse et al.*, 2005; *Prowse et al.,* 2011b]. Numerous studies suggest the utility of lake ice phenology as an index of long-term climate variability and change [e.g., *Robertson et al.,* 1992; *Wynne and Lillesand,* 1993; *Latifovic and Pouliot,* 2007; *Prowse et al.*, 2011b], and in some cases, trends in ice conditions can be considered a more robust measure of local climate than instrumental records [*Livingstone,* 1997]. The strength of ice phenology as a climate proxy include the broad spatial distribution of sites, the high resolution of the data, an often longer record compared to other direct measures such as air temperature, and the relative ease and precision of measuring freezeup and breakup dates both directly and by satellite [*Prowse et al.*, 2011a]. Despite these benefits, historical evaluations of changes in freshwater ice conditions over the circumpolar Arctic have been proven difficult to synthesize because of the limited number of detailed observations [*Prowse et al.*, 2011a]. Long-term records (~150+ years) exist only from a small set of observation sites around the Northern Hemisphere [*Magnuson et al.,* 2000; *Prowse et al.*, 2011c], and unfortunately, these networks have been declining since the mid-1980s. In Canada, for example, the freezeup/breakup network of 2000–2001 only represents 4% of what it was in 1985–1986 [*Lenormand et al.*, 2002]. This has led to serious geographical and temporal gaps for several lake and river ice parameters.

In this study, we will provide new insights on the variation of the long-term ice cover phenology in northern Fennoscandia by using paleolimnology to reconstruct past ice conditions in four small and shallow subarctic lakes. We focus on diatoms that are a highly diverse and abundant group of phytoplankton in the aquatic environment.

The central argument behind the approach is that ice cover in colder regions is one of the strongest single physical parameters affecting lake ecology [*Gabathuler*, 1999] and especially the composition of diatom assemblages in high Arctic lakes [*Keatley et al.*, 2008]. The lengthening of the open water season, especially in subarctic/arctic lakes, increases the overall primary production [*Prowse et al.,* 2006]. Studies of *Regier et al.* [1990] suggest that a 10°C increase in water temperature may lead to a fourfold increase in primary production. Decreased ice cover duration also increases the availability of light, especially in the northern regions due to 24 h daylight in summer and can increase the duration and strength of thermal stratification in lakes. A longer open water period (i.e., longer growing season) can directly affect water column properties such as mixing dynamics, light availability and stability, and nutrient distributions. This has been observed to favor small-celled planktonic diatoms with a high surface area to volume ratio, particularly within the *Cyclotella* genus, as documented by sedimentary studies from the Arctic [*Rühland et al.*, 2008] and monitoring studies from more southerly latitudes [*Winder et al.*, 2009]. The shift from benthic to planktonic dominance, and its links to changing ice cover [*Smol and Douglas,* 2007], has already been shown in arctic and subarctic lakes [*Smol et al.*, 2005]. A decrease in ice cover duration can lead to an increase in aquatic habitat availability, particularly at high latitudes [*Prowse et al.*, 2006]. A comparison of diatom-based paleolimnological profiles from undisturbed lakes across the Northern Hemisphere reported a significantly earlier shift to a more planktonic diatom assemblage in the Arctic than in temperate regions, consistent with this being a response to warming-induced increases in the ice-free period [*Rühland et al.*, 2008]. In years with cold summers and extensive ice cover, diatoms from aerophilous and shallow habitats often dominate even in deep lakes [*Smol*, 1988]. Conversely, during warmer years with less extensive ice cover, taxa characteristics of deeper water substrates and planktonic habitats increase in abundance relative to the shallow water benthic taxa [*Smol*, 1988]. In shallower ponds, warming has been found to increase species richness from an assemblage characterized by a few taxa that compete well in low-sunlight and low-nutrient environments to a more complex assemblage of various benthic taxa that can readily exploit newly developed aquatic habitats [*Smol and Douglas*, 2007].

It is for all these reasons that diatoms, being sensitive to seasonality, water temperature, and water column stability as well as habitat changes, are exceptionally good recorders of lake ice phenology [*Smol and Douglas*, 2007]. Here we develop and use a quantitative diatom-based inference model for the first time to reconstruct the Holocene ice cover history at four subarctic lakes in northwestern Fennoscandia. The melting and freezing degree day approach, on which the quantitative diatom-based model is based on, can be widely applied to relatively small and shallow lakes in the circumpolar region.

2. Study Area

The four study lakes where we track past changes in ice phenology are located across different vegetation and climatic zones in northwestern Fennoscandia. The lakes are located at altitude from 355 to 704 meters above sea level (masl) (Figure 1) and have site-specific mean July air temperatures between 9.7 and 11.4°C. In general, the lakes are small (4–100 ha), relatively shallow (4–22 m), oligotrophic, and circumneutral lakes (pH~7) (Table 1). Their current ice cover duration varies between 225 and 245 days, as inferred using the method described below. As the ice cover duration of these four lakes is in the middle of the ice cover duration gradient of the calibration lake data set used for developing the prediction model (mean: 232; median: 231 days), they are well suited for this study.

3. Methods

3.1. Transfer Model for Hindcasting Past Ice Cover Duration

The calibration data set used to develop a diatom-based transfer model to hindcast past ice cover duration comprises 64 subarctic lakes described by *Weckström and Korhola* [2001] (Figure 1). Ice cover, together with pH (strongest), calcium (Ca), and loss on ignition (LOI), is one of the most important environmental variables explaining the distribution of diatoms in this data set. Despite the relatively low ratio λ_1/λ_2 of 0.45, ice cover has the highest, statistically significant correlation with the DCA (detrended correspondence analysis) axis 2 sample scores ($r = 0.65$, $p < 0.05$). In contrast, pH, Ca, and LOI all show statistically significant correlations with DCA axis 1 sample scores ($r = 0.75$, $p < 0.05$; $r = 0.7$, $p < 0.05$; and $r = -0.7$, $p < 0.05$).

Due to remote locations of the calibration data set lakes, no observed ice cover data are available from the lakes. The ice cover duration data were hence calculated for each lake using the melting degree day value of +130 and a freezing degree day value of −30 based on *Thompson et al.* [2005]. The degree day values were validated by *Thompson et al.* [2005] using three data types: (1) ice cover duration recorded for the period 1961–1990 from 21 Finnish lakes, (2) thermistor chain data collected during the European-wide European Mountain lake Ecosystems: Regionalisation, Diagnostics, and Socioeconomic Evaluation project in the winter 2000–2001 for a small number of lakes from each of the lake regions, and (3) a 162 year long historical record of ice cover from Lake Kallavesi, Central Finland. Moreover, we further tested the robustness of the degree

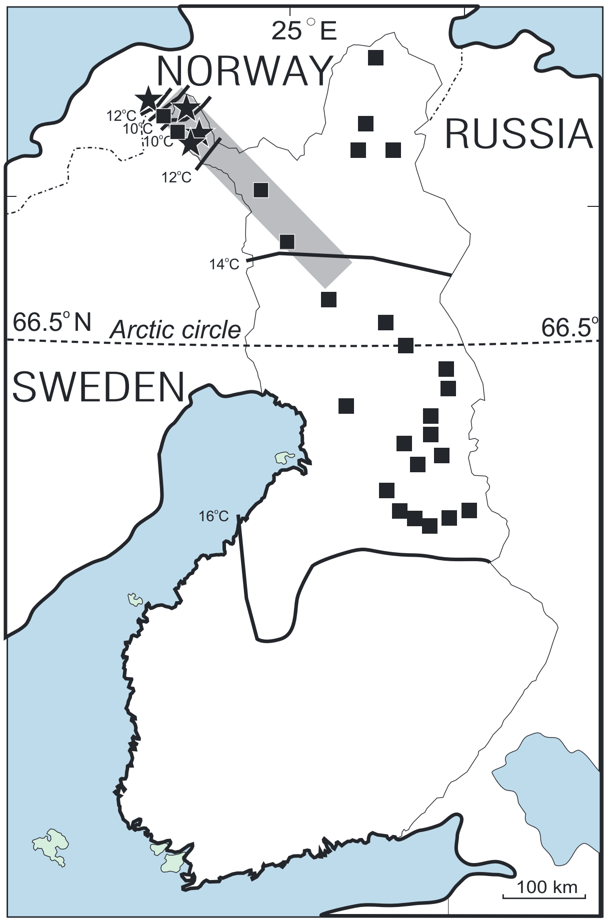

Figure 1. Locations of the four long-term study lakes (asterisk), the 64 calibration data set lakes (gray rectangle), and the 25 degree day model test set lakes (filled squares). The normal period 1981–2010 July mean temperature isotherms are shown as black lines [*Pirinen et al.,* 2012].

day model by comparing the degree day inferred ice cover duration and the mean observed ice cover duration between 1961 and 2000 of 25 northern Finnish lakes varying in size and depth; the strong correlation ($r = 0.96$, $p < 0.05$) further demonstrates the potential of this model. The inferred ice cover length for the 64 calibration data set lakes varies from 204 to 262 days.

A gravity corer was used to collect surficial sediment from the deepest part of the 64 calibration lakes, and diatoms were analyzed from the topmost 1 cm thick sediment sample representing the modern communities. Due to the different sediment accumulation rates in lakes located in different vegetation/climatic zones, and thus a different temporal extent of the topmost sediment sample, daily temperature data of 10, 20, and 30 years were used for sites located in the coniferous forest, mountain birch woodland, and barren tundra, respectively. The cutoff values are based on the previously assessed sedimentation rates of lakes within this region [*Korhola and Weckström*, 2004]. The daily temperature for each site was derived from a set of climate stations [*Drebs et al.*, 2002] using universal kriging [*Olea*, 1974]. The method interpolates a parameter field by taking into account the spatial covariance and distance to samples. The impact of a single station to the inferred temperature of a site increases as the distance between them decreases. We estimated the variogram (spatial covariance of error) by a linear model. The intercept was fixed to zero, and the effect was estimated from the stations' temperature data by calculating the sample variance of the temperature difference with respect to distance between stations. The distance is calculated by the Euclidean distance between coordinates, approximating the surface by a flat plane. The effect was 3.13 with a square correlation of 0.78 ($p < 10^{-9}$). Universal kriging further calculates the general trend in the field by assuming a user-defined shape for it. We used first-degree polynomials as the shape of the trend, as temperatures decrease when moving north. The daily temperatures for each site were calculated by first lowering all the stations to the sea level with a lapse rate correction of 0.53°C/100 m [*Laaksonen*, 1976]. Then, universal kriging was applied to calculate the daily temperatures from

Table 1. Locations and Environmental Data of the Four Studied Lakes

Variable	Lake Dalmutlad´do	Lake Vallijärvi	Lake Tsuolbmajavri	Lake Toskaljavri
Latitude (°N)	69.18	68.68	68.68	69.19
Longitude (°E)	20.72	21.58	22.05	21.45
Altitude (masl)	355	457	526	704
Area (ha)	7.21	3.60	13.91	100.35
Maximum depth (m)	3.40	5.20	5.35	21.50
Inferred ice cover duration (days)	225	228	232	245
pH (units)	6.9	6.8	7.4	7.2

Table 2. Radiocarbon ^{14}C AMS Measurements and Calibrated Ages for Lake Vallijärvi

Laboratory No.	Depth (cm)	Radiocarbon Age (^{14}C year B.P.)	Calibrated Mean 1σ Age (B.P.)
Poz-4798	25–26	1690 ± 30	1590
Poz-4799	54–55	2520 ± 30	2590
Poz-4801	63–64	2870 ± 80	3010
Poz-4802	81–82	3590 ± 40	3900
Poz-4803	100–101	4915 ± 35	5640
Poz-4804	134–135	7660 ± 50	8460
Poz-4805	162–163	7790 ± 70	8570
Poz-4807	173–174	8210 ± 50	9180
Poz-4808	181–182	8660 ± 50	9620
Poz-4809	192–193	7420 ± 50	8260

the stations, and the inferred temperatures were lapse rate corrected back to the site's altitude. The degree days were then calculated for each site and each year from the inferred temperature data, and the average ice cover duration was calculated over 10, 20, or 30 years, depending on the estimated temporal extent of the sediment sample.

The diatom-based transfer function for inferring past ice cover durations was established using a two-component weighted averaging partial least squares [*ter Braak and Juggins*, 1993] model yielding a leave-one-out cross-validated root-mean-square error of prediction of 7.2 days and r^2 of 0.66 between the observed and diatom-inferred values. As the four reconstructed lakes are part of the original calibration set, each of them was removed by turns during modeling, in order to ensure the independency between the calibration set lakes and the reconstructed lakes. The modeling procedures were performed using the program C2 [*Juggins*, 2007].

3.2. Sediment Cores

The sediment cores covering the Holocene were recovered from lakes Tsuolbmajavri, Toskaljavri, and Dalmutlad´do in May 1997, April 1999, and April 2001, respectively, and their chronologies are based on 14, 8, and 11 accelerator mass spectrometry (AMS) ^{14}C dates, respectively [*Seppä and Weckström*, 1999; *Seppä et al.*, 2002; *Nyman et al.*, 2008]. The sediment samples from Lake Vallijärvi were retrieved in April 2003, dated at the Poznan Radiocarbon Laboratory, Poland, and the chronology is based on 10 AMS ^{14}C dates (Table 2). The dates are mainly based on terrestrial wood fragments and plant remains (e.g., seeds, leaves, and moss stems). The radiocarbon dates were converted to calendar years using the software Calib 5.1 (IntCal04) [*Stuiver and Reimer*, 1993; *Reimer et al.,* 2004].

4. Results and Discussion

4.1. Long-Term Ice Cover History

The results of the ice cover inferences are shown as anomalies from the Holocene mean (Figure 2). All four lakes show generally similar features in their ice cover duration, although distinctive differences also occur. Lakes Tsuolbmajavri and Vallijärvi correspond rather well with the overall regional climate history of the area reconstructed independently by means of several pollen records [*Weckström et al.*, 2010], whereas Lakes Dalmutlad´do and Toskaljavri show some deviations. This might be due to the different location of the sites in relation to the temperature/altitude gradient [e.g., *Hodgins et al.*, 2002; *Weyhenmeyer et al.*, 2004]. Lake Dalmutlad´do is situated on the western flank of the Scandes and is thus influenced by a more marine climate, Lake Toskaljavri is located on barren tundra, and Lakes Tsuolbmajvri and Vallijärvi lie at the treeline of pine (*Pinus sylvestris*). The different responses of these lakes might also be caused by different biological interactions within the lakes [*Carpenter et al.*, 1987], or the variable watershed characteristics and topographical impacts [e.g., *Bigler et al.*, 2003; *Leavitt et al.*, 2009], as lakes and their catchments are essentially individuals, and even closely located lakes may respond differently to environmental changes [e.g., *Keatley et al.*, 2008]. However, all reconstructions show mainly higher than Holocene average ice-free periods for the early Holocene (approximately 10 to 6.5 calibrated kyr (cal kyr) B.P.) and shorter than average ice-free periods since 6.5 kyr, culminating during the so-called Little Ice Age (approximately 500 to 100 calibrated yr B.P.), especially in Lake Vallijärvi. During the early Holocene, where the pollen-inferred mean July air temperature was estimated to be approximately 1°C warmer than the Holocene mean, the ice cover duration was as much as 1 week shorter than the Holocene mean, whereas during the Little Ice Age, the ice cover duration was between 3 and 5 days longer than the Holocene mean. The impact of the recent climate warming is reflected as prolonged ice-free periods.

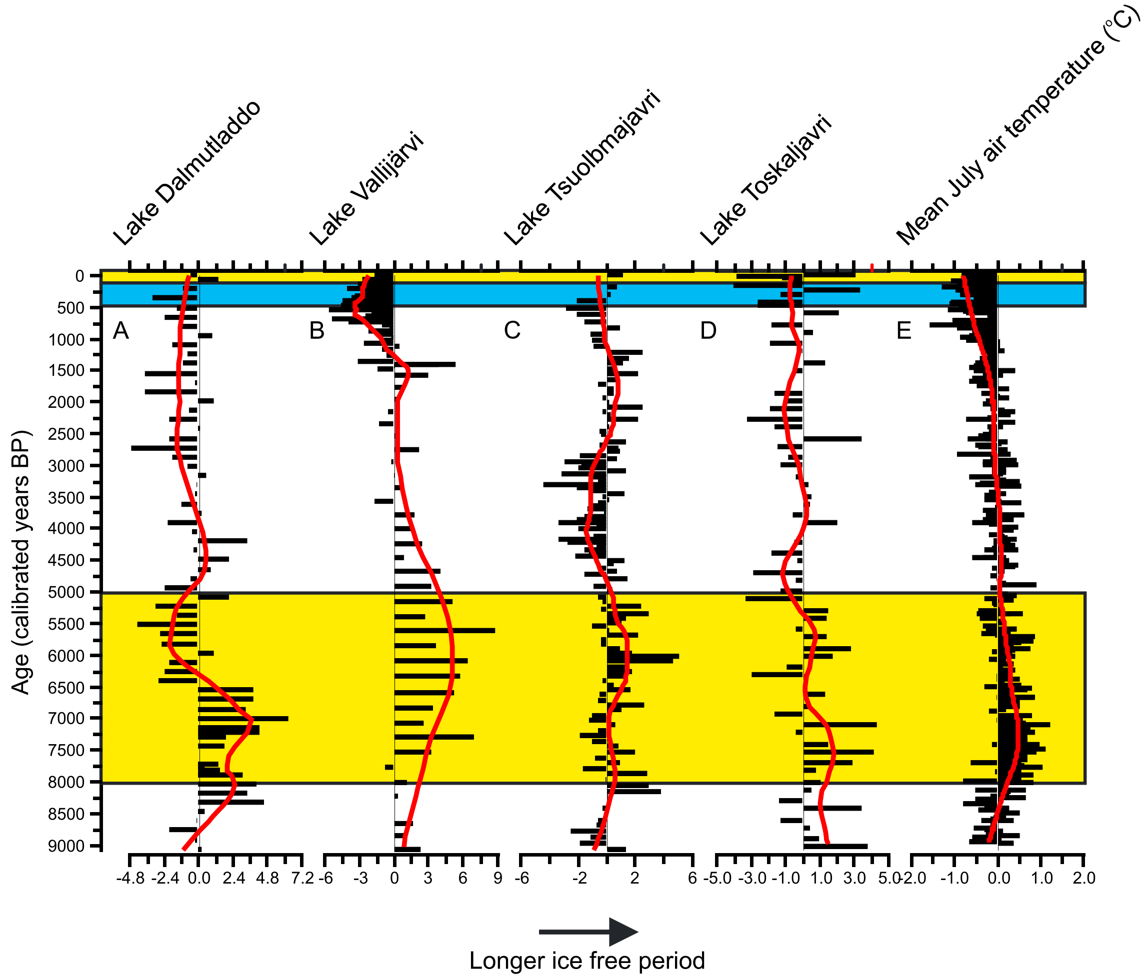

Figure 2. (a–d) Diatom-based Holocene ice cover phenology of four subarctic lakes and (e) an independent regional pollen-based July mean air temperature reconstruction. The ice cover values (days) are deviations from the Holocene mean ice cover duration and the pollen values (°C) deviations from the Holocene mean July air temperatures. A locally weighted scatterplot smoother (lowess; span 0.2) is added to highlight the general trends. The warmer periods (Holocene thermal maximum and the recent climate warming) are indicated with a yellow box, and the cooler period of the Little Ice Age is indicated with a blue box.

The changes in the diatom assemblages occur mainly within the benthic taxa as all lakes except Toskaljavri are relatively shallow (maximum depth <5.5 m). Although planktonic diatoms are common in the sediments of the deeper lake Toskaljavri, changes were not similar to the shifts between benthic and planktonic taxa reported by *Smol et al.* [2005] and *Smol and Douglas* [2007].

4.2. General Discussion

The use of transfer functions has lately been under critical review, and scientists are urged to be more cautious when applying such models [*Juggins*, 2013]. Although ice cover was not found to be the strongest environmental variable affecting the diatom composition in this training set, it was however found to be ecologically important having the strongest correlation of all variables to DCA axis 2 sample scores. Unlike in the Alps [see *Psenner and Schmidt*, 1992], there is no statistically significant correlation between pH and ice cover ($r = 0.11$, $p = 0.392$) in northwestern Fennoscandia. Moreover, as many of these lakes have experienced a subtle gradual long-term acidification during the Holocene [e.g., *Korhola and Weckström*, 2004; *Nyman et al.*, 2008], it is plausible to assume that the main changes in the diatom assemblages are caused by changes in the ice cover or air temperature (the correlation between air temperature and ice cover in this data set is -0.98, $p < 0.05$).

Lake size and depth have a strong effect on the timing of freezeup, owing to the heat content of large water bodies [*Smol and Douglas*, 2007]. Mean lake depth is relatively well correlated with freezeup dates due to the capacity of deep lakes to store more heat than shallower lakes. Ice breakup, on the contrary, is more dependent on the surface area of the lake due to the wind-induced mechanical pressure [*Palecki and Barry*, 1986]. The melting and freezing degree day approach used in this study however can provide a powerful means to reconstruct and/or to predict the ice cover duration of medium sized, relatively shallow lakes (see below).

Although the formation and melting of ice cover are a function of multiple environmental factors and their interaction (e.g., climate variables, shading, lake size, and depth), our results suggest that relatively accurate, robust, and effective ice cover models can be developed. The calculated melting degree day values of +130 and a freezing degree day value of −30 for each lake [see *Thompson et al.*, 2005] used in this study perform well in relatively small, moderately deep high-latitude lakes. The robustness of the degree day approach has also been shown by, e.g., *Ruosteenoja* [1986], who suggested a melting degree day value of +140 days for a large Finnish Lake and by *Agusti-Panareda and Thompson* [2002], who used the degree day value of 120 days for two large Finnish lakes.

The reliability of our diatom-based model reconstructions (1°C increase in air temperature decreases ice cover duration by approximately 7 days) is in good agreement with previous studies. *Loader et al.* [2011] suggested that a decrease in ice cover duration of 14 days in river Torniojoki, Finland, is equivalent to an increase in April and May mean temperature of approximately 2.5°C. Furthermore, *Robertson et al.* [1992] showed that in Lake Mendota, USA, an increase in air temperature by 1°C shortened the ice cover duration by 11 days. This is also supported by *Williams et al.* [2004], who studied 143 North American lakes and also suggested that a 1°C increase in air temperature reduces the ice cover duration by approximately 11 days.

Although the impact of increased temperature on the length of ice cover is very strong, there are considerable differences in the magnitude of change of lakes located in climatologically different areas. In Sweden, for example, *Weyhenmeyer et al.* [2004] have demonstrated that an increase in air temperature by 1°C causes ice breakup to occur approximately 17 days earlier in warmer regions in southern Sweden but only 4 days earlier in the cooler northern parts. The same phenomenon has also been shown by *Hodgins et al.* [2002] in New England, USA, where the temperature increase of 1.5°C during the last 150 years has shifted the ice breakup to occur 16 days earlier in the south, whereas only 9 days earlier in the north. This is further supported by *Beier et al.* [2012], who have documented a rapidly decreasing trend of up to 21 days less ice cover between 1975 and 2007 in five lakes located in the Adirondack Mountains, northern New York state, USA, close to the southern New England lakes. As the length of ice cover determines the timing and nature of many physical, chemical, and biological processes, the shift in timing of the ice breakup of lakes located along a latitudinal temperature gradient may change the diversity of lake types in many areas in the future [*Weyhenmeyer et al.*, 2004].

5. Conclusions

A diatom-based ice cover model was developed to quantitatively infer Holocene ice cover histories in four small lakes from subarctic Fennoscandia. The reconstructed ice cover histories had similar features with independently reconstructed regional climate history. Our data show that ice-free periods during the warmer early Holocene (approximately 10 to 6.5 cal kyr B.P.) were generally longer than the Holocene average and shorter than the Holocene average ice-free periods since 6.5 kyr toward present, with mainly longer ice cover durations during the Little Ice Age (approximately 500 to 100 cal yr B.P.). When interpreting the results of lacustrine biological proxies, local environmental and site-specific conditions should be considered as well as potential challenges related to quantitative reconstructions. Diatom-based ice cover inference models can provide a powerful tool to reconstruct past ice cover history in remote and sensitive areas where no historical measurements are available.

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